BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary Addition and Shifts

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Recall · Lesson 001

Recall Lesson 001 before reading the explanations. Answer all four questions.

1 · Previous lesson. Convert 00101101 to denary. Show the place values you use.

2 · Multiple choice. Why do computers use binary?
A Ten states per bit · B Two reliable states · C No codes · D Only numbers

3 · Earlier learning. Convert denary 150 to unsigned eight-bit binary.

4 · Thinking. Explain why the highest unsigned eight-bit value is 255, not 256.

Today's objectives

Add eight-bit binary numbers including carries; recognise overflow; perform logical shifts and explain what happens to discarded bits.

Key vocabulary · read, cover, recall

carry
Moved to next left column

overflow
Result needs extra bits

unsigned
No negative values

logical shift
Move bits; fill zeros

Write: What does overflow mean in an eight-bit unsigned calculation?

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Learn & follow

The rules of binary addition

0 + 0 = 0 · 0 + 1 = 1 · 1 + 0 = 1 · 1 + 1 = 10_2 (write 0, carry 1).
1 + 1 + carried 1 = 11_2 (write 1, carry 1). Start at the right-hand bit.

I Do · Worked example

Model: Add 00101101 (45) and 00010111 (23). The green carry row shows the 1 entering the next column.

Carry in	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**0**
First number	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**1**
+ Second	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**1**
= Result	**0**	**1**	**0**	**0**	**0**	**1**	**0**	**0**

1. Start at the right: 1 + 1 = 10_2. Write 0, carry 1 to the next column.
2. Move left: add each pair of bits and any carry above them.
3. Finish and check: 01000100 = 68, and 45 + 23 = 68.

Overflow · when the answer needs nine bits

255 + 1 = 256 = 1 00000000_2. An unsigned eight-bit answer has room for only eight bits. The extra ninth bit means overflow.

Logical left shift · see what moves

Before: 00010110 = 22. Move every bit two columns left, then put zeros in the empty places on the right.

BEFORE	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**0**
MOVE LEFT 2				←				
AFTER	**0**	**1**	**0**	**1**	**1**	**0**	**0**	**0**

After: 01011000 = 88. Bits beyond the left edge are lost (discarded).

We Do · Guided practice

1. Add 00001101 + 00000111. Write carry digits above; give the result and denary value.

Carry									
Number 1		**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**
+ Number 2		**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**
= Answer	9th								

2. Shift 00101000 one place right. Write the new eight-bit pattern and denary value.

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Think & write

Code First · Predict → Write → Test → Improve

Predict: What will print(11 + 6) display? Write the result.

Write: Write a print(...) instruction to add 13 and 7.

Test on paper: Work out 13 + 7. Does the answer match what your code would display?

Improve: Change your instruction to add 38 and 27. What will it display?

You Do · Exam-style practice

1. Add 00001011 + 00000110. Write carries and the eight-bit result.

Carry									
Number 1		**0**	**0**	**0**	**0**	**1**	**0**	**1**	**1**
+ Number 2		**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**
= Answer	9th								

2. Add 00110101 + 00010110. Show the carries; check in denary.

Carry									
Number 1		**0**	**0**	**1**	**1**	**0**	**1**	**0**	**1**
+ Number 2		**0**	**0**	**0**	**1**	**0**	**1**	**1**	**0**
= Answer	9th								

3. Add 11110000 + 00100000. Put any ninth bit into the overflow box.

Carry									
Number 1		**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**
+ Number 2		**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**
= Answer	9th								

Explain: Has overflow occurred? Why?

4. Logical left shift 00001011 two places. Give the eight-bit result and denary value.

Thinking: Why does shifting left not always multiply the value when a bit is lost?

Exit Ticket · answer before turning the page

Choose True (T) or False (F). Circle one answer for each question.

1. In binary, 1 + 1 gives 10_2. ☐ T ☐ F
2. All unsigned eight-bit sums fit in eight bits. ☐ T ☐ F
3. A logical right shift puts zeros in the empty left positions. ☐ T ☐ F

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Review & improve

Now switch to GREEN PEN. Find your answer on pages 1–3, check it, and tick if correct. If not, use the correct answer to learn what to change.

Page 1 · Do Now

Q1	45 · 32 + 8 + 4 + 1 = 45 (the 1-bits).	☐
Q2	B — two reliable states · Binary uses two states, 0 and 1.	☐
Q3	10010110 · 128 + 16 + 4 + 2 = 150.	☐
Q4	255 · Eight bits make 256 values: 0–255.	☐

Page 2 · We Do

Q1	00010100 = 20 · 13 + 7 = 20; carry across columns.	☐
Q2	00010100 = 20 · A right shift adds 0 at the left.	☐

Page 3 · Code First

Predict	17 · 11 + 6 = 17.	☐
Write / Test	print(13 + 7) → 20 · The printed output is 20.	☐
Improve	print(38 + 27) → 65	☐

Page 3 · You Do

Q1	00010001 = 17 · 11 + 6 = 17.	☐
Q2	01001011 = 75 · 53 + 22 = 75.	☐
Q3	1 00010000 = 272 · Overflow: the extra ninth bit cannot fit.	☐
Q4	00101100 = 44 · 11 × 4 = 44; no bits lost.	☐
Thinking	A bit can be discarded. · The result then loses that bit’s value.	☐

Page 3 · Exit Ticket

Q1	True · 1 + 1 = 10_2. Write 0; carry 1.	☐
Q2	False · Some answers need nine bits.	☐
Q3	True · Zero enters at the left edge.	☐

Try the question again

Choose one you got wrong. All correct? Choose the hardest one.

1. Copy the whole question:

2. Write a full new answer in GREEN PEN. Show any working: